

WARA Controller

Runs gates, freezes contracts, and routes repair

commit / read

Wireless Artifact Workspace

Stores problems, models, code, logs, figures, references, and claims

Phase 1 Research gap identification and problem proposal

Subphase 1.1
Identification of
Wireless System,
Variables, and Metrics
ScoutAgent

topic

Subphase 1.2
Literature Search
LiteratureAgent

literature

Subphase 1.3
Generation and Selection
of Research Problems
ScoutAgent

problem

Subphase 1.4
Freezing and Exporting
the Selected Problem
LiteratureAgent

problem handoff

Phase 2 Wireless optimization modeling, algorithm design, and experimentation

Subphase 2.1
Construction of
System Model and Optimization
Problem
FormulationAgent

model

Subphase 2.2
Tractability Analysis and
Solution-Route Selection
TheoryAgent

route

Subphase 2.3
Algorithm
Specification and
Validation Principle
TheoryAgent

algorithm

Subphase 2.4
Experiment Design
and Code Execution
ExperimentAgent

quick run

Subphase 2.5
Result Verification
and Promotion
to Paper-Level Evidence
ValidationAgent

math contract

algorithm contract

executable package

verified result package

verified results

experiment / result repair

result

Phase 3 Research deliverable generation

Subphase 3.1
Drafting Problem
Formulation and Method Sections
WritingAgent

tech

Subphase 3.2
Drafting
Numerical Results
and Supported Claims
AnalysisAgent

claims

Subphase 3.3
Assembly of
Technical Sections
WritingAgent

sections

Subphase 3.4
Construction of the Full
Manuscript Draft
LiteratureAgent
+ WritingAgent

citations

Subphase 3.5
Full Manuscript Review
ReviewAgent

final

Subphase 3.6
Revision and
Export of the
Final Research Package
RepairAgent +
WritingAgent

manuscript repair



main artifact flow



phase handoff



localized repair

frozen contract